



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to LR AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

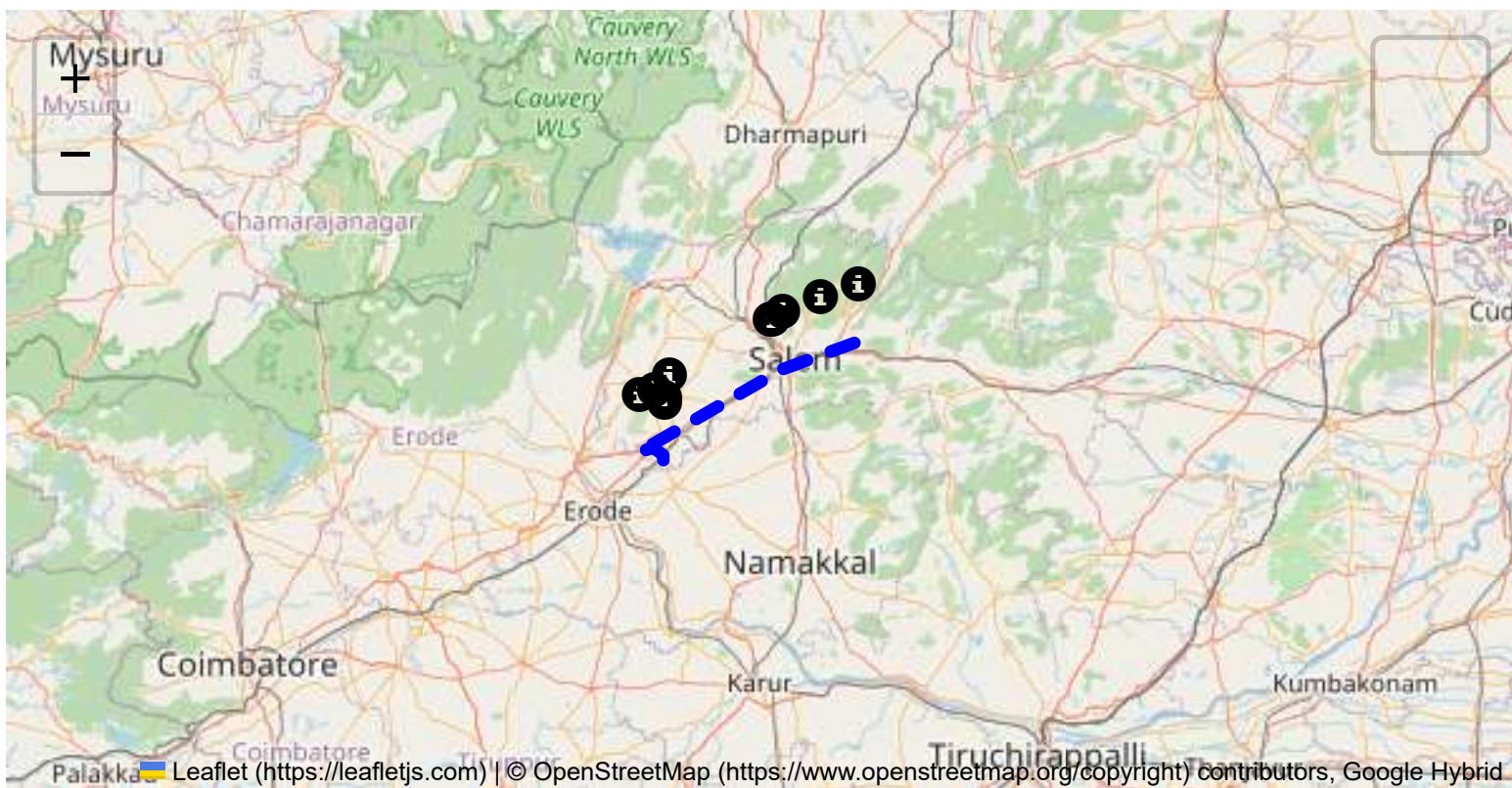
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

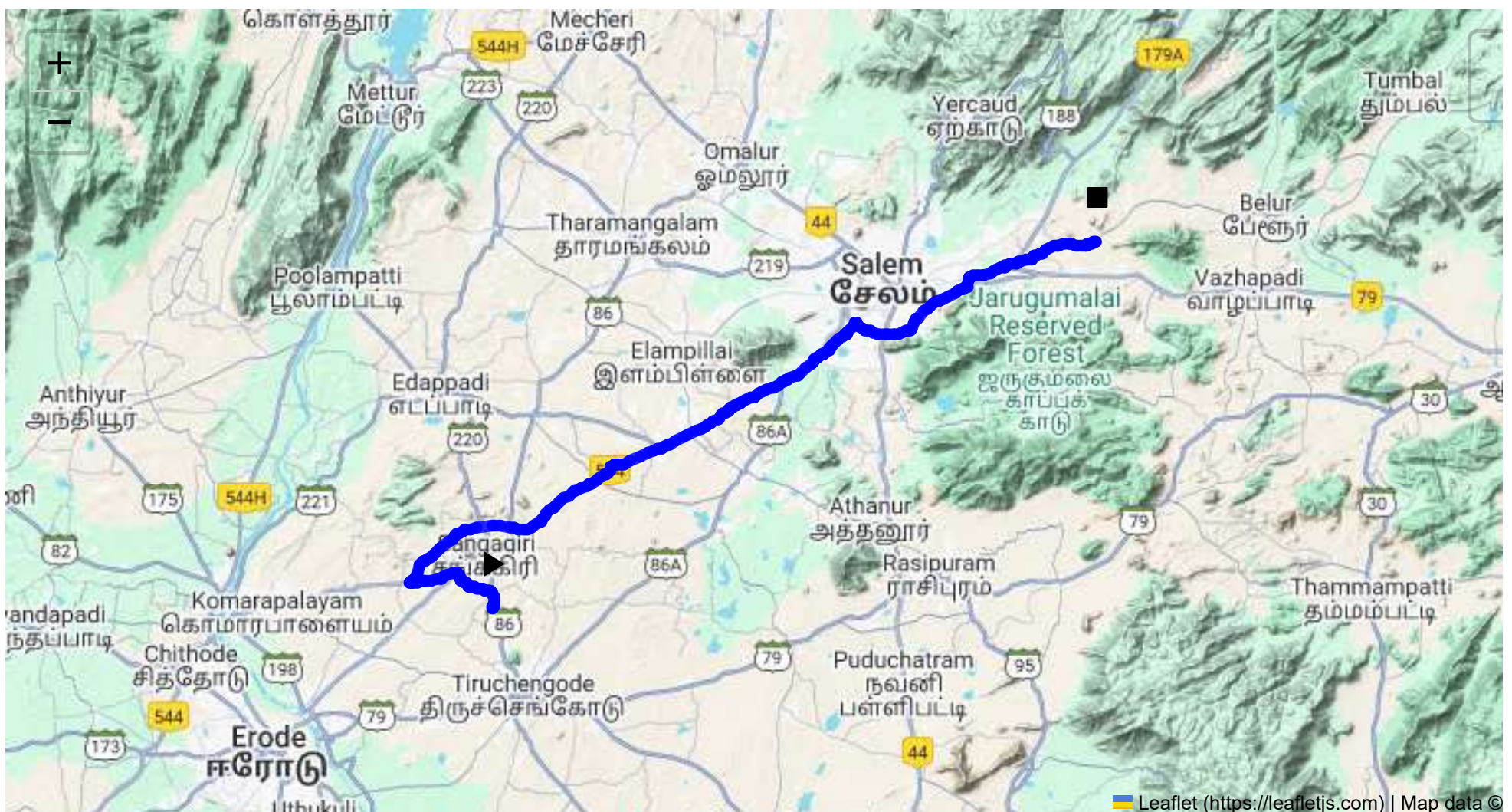
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 70.29 km
Estimated Duration: 1.5 hours
Adjusted Duration (Heavy Vehicle): 1.9 hours
Start: (11.4381, 77.8734)
End: (11.68903357, 78.29473314)



Welcome to the Journey Risk Management Study

Here's a comprehensive analysis of the route from Sangagiri to Natamangalam, including its potential hazards and considerations for truck drivers carrying hazardous materials:

1. Overview of the Route Map:

- The route spans approximately 70.29 kilometers through Tamil Nadu, primarily along regional and national highways. It passes through several towns and cities including Sangari RS, Salem,

and Udayapatti before reaching Natamangalam.

2. Typical Weather Conditions and Potential Weather-Related Hazards:

- Tamil Nadu experiences a tropical climate with heavy monsoons from June to September, which can lead to waterlogging and reduced visibility. Summers can be particularly hot, which might affect vehicle performance.

3. Traffic Patterns:

- Congestion is common in Salem during morning (8-10 AM) and evening (5-8 PM) peak hours. Especially around the areas of Sankari RS and Salem, due to local business activities and commuting traffic.

4. Assessment of Road Quality and Infrastructure:

- Paved roads with frequent maintenance. However, expect narrow sections, especially near Sankari RS. NH544 is generally well-maintained with multiple lanes but can witness occasional potholes.

5. Suggestions for Alternative Routes for Emergencies:

- In case of emergencies around Salem, bypass routes through either the SH188 or local roads like the Masinaickenpatti bypass can be considered to avoid congestion in central areas.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

- Tamil Nadu has specific routes designated for hazardous goods; ensure compliance with regional and state transportation mandates, including obtaining permits when necessary.

7. Overview of Historical Incidents:

- There have been occasional reported incidents involving heavy vehicles on NH544, typically due to speeding or mechanical failure, though these are infrequent.

8. Environmental Considerations and Sensitive Areas:

- The route cuts through several semi-urban and rural areas. It's important to minimize noise and avoid the release of any pollutants during trucking operations, particularly in populated locales.

9. Analysis of Communication Coverage:

- Mobile coverage is generally good along the entire route, but there might be occasional dead zones in rural segments or near dense foliage, like areas around the outskirts of Salem.

10. Estimated Emergency Response Times:

- Emergency services in towns like Sankari or Salem are accessible, with response times around 15-30 minutes in urban areas, slightly longer in remote sections.

11. An Overall Summary of Risk Assessment:

- The primary risks include potential congestion in Salem, weather-related hazards during monsoon and summer months, and road maintenance issues in rural segments. Familiarity with alternative routes and regular updates on weather and traffic can mitigate most operational risks.

By considering these factors, you can ensure a safer journey for truck drivers transporting hazardous materials along this route. Adhering to local regulations, maintaining awareness of changing weather

conditions, and being prepared for emergencies will be crucial.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63266, 78.12699	15 KM/Hr
1	Turn	Medium	11.43822, 77.87348	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.44900, 77.87406	30 KM/Hr
5	Turn	Medium	11.45357, 77.85789	30 KM/Hr
6	Turn	Medium	11.45352, 77.85698	30 KM/Hr
7	Turn	Medium	11.45847, 77.85140	30 KM/Hr
8	Turn	Medium	11.46048, 77.85036	30 KM/Hr
9	Turn	Medium	11.46303, 77.84945	30 KM/Hr
10	Turn	Medium	11.46314, 77.84928	30 KM/Hr
11	Turn	High	11.45542, 77.81725	15 KM/Hr
12	Turn	High	11.45634, 77.81673	15 KM/Hr
13	Turn	High	11.61233, 78.10397	15 KM/Hr
14	Turn	High	11.61216, 78.10423	15 KM/Hr
15	Turn	High	11.63300, 78.12746	15 KM/Hr
16	Turn	Medium	11.65627, 78.20215	30 KM/Hr
17	Turn	Medium	11.65664, 78.20240	30 KM/Hr
18	Turn	High	11.66380, 78.20648	15 KM/Hr
19	Turn	Medium	11.67128, 78.22971	30 KM/Hr
20	Turn	High	11.67111, 78.23134	15 KM/Hr
21	Turn	Medium	11.67195, 78.23146	30 KM/Hr
22	Turn	Medium	11.67666, 78.24906	30 KM/Hr
23	Turn	Medium	11.68494, 78.26198	30 KM/Hr
24	Turn	Medium	11.68551, 78.26771	30 KM/Hr
25	Turn	Medium	11.68624, 78.28833	30 KM/Hr
26	Turn	Medium	11.68848, 78.29399	30 KM/Hr
27	Turn	High	11.68904, 78.29460	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
9	hospital	Vinayaka Mission's Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
13	hospital	Thayammal Hospitals	11.6237472, 78.1177631	30 km/h	Medium
14	hospital	Vanitha Hospital	11.6239931, 78.1184622	30 km/h	Medium
16	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
17	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
18	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
19	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
20	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
21	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
22	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
23	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
24	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
25	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
26	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
8	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
10	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
11	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
12	school	Sri Vidhya Bharati Matric HR Sec School	11.6186069, 78.1103239	30 km/h	Medium
15	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
27	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
28	school	Govt Hr Sec School	11.6754733, 78.2459724	30 km/h	Medium
29	school	St Mary's Hr Sec School	11.6872096, 78.2796818	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63266, 78.12699



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45357, 77.85789



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45847, 77.85140



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46048, 77.85036



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46314, 77.84928



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45634, 77.81673

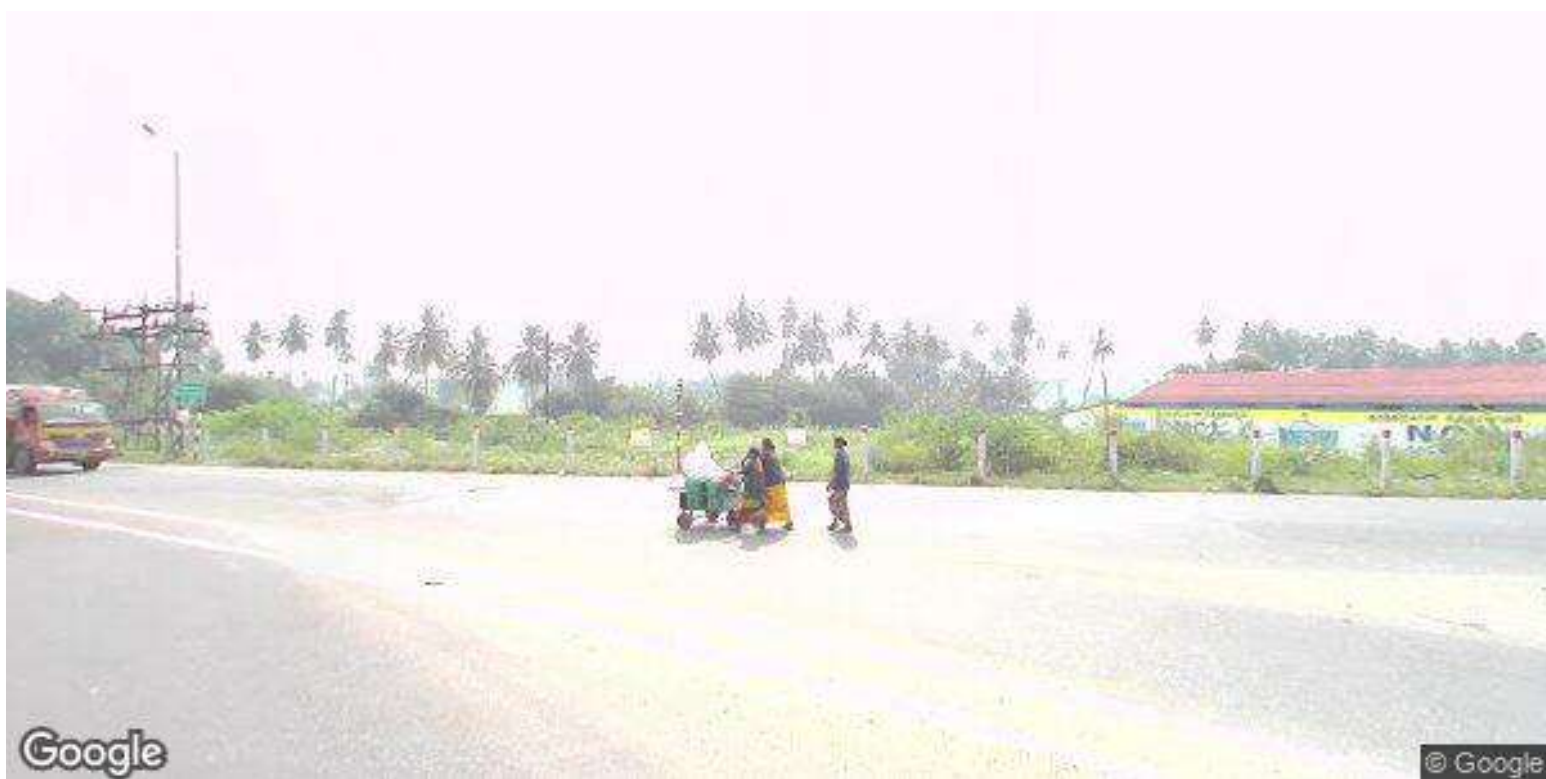


Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.61233, 78.10397



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.61216, 78.10423



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.63300, 78.12746



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65627, 78.20215



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.65664, 78.20240



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.66380, 78.20648



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67128, 78.22971



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67111, 78.23134



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67195, 78.23146



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67666, 78.24906



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.68494, 78.26198



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68551, 78.26771



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68624, 78.28833



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68848, 78.29399



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.68904, 78.29460
